CURRICULUM VITAE

Zikai Zhang（张子凯）

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EDUCATION

Jan. 2023-Present **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Ph.D. in Computer Science and Engineering

Sep. 2018-Jun. 2021 **Department of Computer Science and Technology, Huaqiao University, China**
M.Eng. in Computer Technology

Sep. 2014-Jul. 2018 **Department of Mechatronics and Vehicle Engineering, East China Jiaotong University, China**
B.Eng. in Mechanical Manufacturing and Automation

RESEARCH INTEREST

Machine Learning, Efficienct AI, Security, Large Language Model, Edge and Cloud Computing

TEACHING INTEREST

Artificial Intelligence, Computer Programming, Cybersecurity, Computer Vision, Communication Networks, Cloud Computing

RESEARCH EXPERIENCE

Jan. 2023-Present **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Graduate Research/Teaching Assistant, Advisor: Dr. Rui (Zoey) Hu

Sep. 2022-Nov. 2022 **Institute of Perceptual Interaction, Nanchang Virtual Reality Research Co., Ltd, China**
Algorithm Engineer

Aug. 2021-Mar. 2022 **Institute of Intelligent Vehicle, Tongji University, China**
Research Assistant

Apr. 2021-Jul. 2021 **Department of Research, DMAI, Inc., Guangzhou, China**
Research Intern

Apr. 2018-Jun. 2021 **Computer Vision and Pattern Recognition Lab, Department of Computer Science and Technology, Huaqiao University, China**
Advisor: Dr. Bineng Zhong
M.Eng Thesis: Deep Convolutional Neural Network based Long-term Object Tracking and Object Detection

MENTORING/TEACHING EXPERIENCE

Mentoring:

2023-2024 **Mentor,** Capstone Project for Undergraduates, **University of Nevada, Reno, Reno, Nevada, USA.**
- Students: Cody Long, Zachary Strazi, Kristian Konstantinov, Jacob Ayers
- Project: Automatic Speech Recognition Attack with Adversarial Examples

2023 Summer **Mentor,** Research Experiences for Undergraduates (REU), **University of Nevada, Reno, Reno, Nevada, USA.**
- Students List: Manny Ortiz (Penn State University), Zhang Lin (Colorado School of Mines)
- Project: Backdoor Attack in Federated Learning System
- Supported by NSF #2150394

Teaching:

2025 Fall	**Graduate Teaching Assistant,** CS442/642 Cloud Computing, **UNR, Reno, Nevada, USA.**
2025 Spring	**Graduate Teaching Assistant,** CS302 Data Structure, **UNR, Reno, Nevada, USA.**
2019-2020	**Sessional Lecturer,** Web Interaction Design **for Undergraduates in Software Engineering Department, Xiamen Institute of Technology, Xiamen, Fujian, China.**

- Introduction of JavaScript; JavaScript Objects and Arrays; Introduction of jQuery; jQuery Selector; jQuery DOM Object; jQuery Elements and Sets; jQuery Events and Animation; jQuery Function; Validation Engine and Regular Expression.

AWARD AND GRANT

Oct. 2025	**First Place Poster,** 2025 UNR CyberSecurity Conference, Reno, NV, USA.
Aug. 2025	**HDRFS Student Publication & Travel Support,** NSF EPSCoR, USA.
Oct. 2024	**First Place Poster,** 2024 UNR CyberSecurity Conference, Reno, NV, USA.
2024-2025	**Co-PI, Nevada WateReuse Research Grant,** WateReuse Association**,** USA.
Oct. 2023	**2023 IEEE CNS Travel Grant,** NSF, USA.
Oct. 2023	**First Place Poster,** 2023 UNR CyberSecurity Conference, Reno, NV, USA.
Oct. 2020	**TOP 1% (5/489)**, **Electric Vehicle Helmet Recognition Competition**, The 3rd Chinese Conference on
Aug. 2020	Pattern Recognition and Computer Vision (PRCV 2020), Nanjing, China.
	TOP 1% (22/2245), Silver Medal, **Kaggle Global Wheat Detection Competition**, Computer Vision Problems in Plant Phenotyping (CVPPP) Workshop, 2020 European Conference on Computer Vision (ECCV), Online.
2018-2021:	**Sole PI, Graduate Innovation Fund** in Scientific Research, Huaqiao University, China.

PUBLICATION & PATENT

Papers in Refereed Journals and Conferences:

[1] **Zikai Zhang**, Rui Hu, Zhiyuan Yu, and Olivera Kotevska. *"STJail: Practical Video Jailbreak Attack Against Multimodel Large Language Models."* Under Review.

[2] **Zikai Zhang**, Rui Hu, and Jiahao Xu. *"SelfGrader: Detecting Jailbreak Attacks on Large Language Models with Token-Level Logit Distribution."* Under Review.

[3] **Zikai Zhang**, Rui Hu, Ping Liu, and Jiahao Xu. *"Fed-pilot: Optimizing LoRA Allocation for Efficient Federated Fine-Tuning with Heterogeneous Clients."* Under Review.

[4] **Zikai Zhang**, Rui Hu, and Jiaohao Xu. *"Heterogeneous Federated Fine-Tuning with Parallel One-Rank Adaptation."* The Fourteenth International Conference on Learning Representations, ICLR (2026).

[5] Jiaohao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**. *"Traceable Black-Box Watermarks For Federated Learning."* The Fourteenth International Conference on Learning Representations, ICLR (2026).

[6] **Zikai Zhang**, Suman Rath, Jiaohao Xu, and Tingsong Xiao. *"Federated Learning for Smart Grid: A Survey on Applications and Potential Vulnerabilities."* ACM Transactions on Cyber-Physical Systems, ACM TCPS (2026).

[7] Yan Gao, Massimo Roberto Scamarcia, Javier Fernandez-Marques, Mohammad Naseri, Chong Shen Ng, Dimitris Stripelis, Zexi Li, Tao Shen, Jiamu Bai, Daoyuan Chen, **Zikai Zhang**, Rui Hu, InSeo Song, Lee KangYoon, Hong Jia, Ting Dang, Junyan Wang, Zheyuan Liu, Daniel Janes Beutel, Lingjuan Lyu, and Nicholas D Lane. *"FlowerTune: A Cross-Domain Benchmark for Federated Fine-Tuning of Large Language Models."* Advances in Neural Information Processing Systems, NeurIPS (2025).

[8] Jiaohao Xu, **Zikai Zhang**, and Rui Hu. *"On the Out-of-Distribution Backdoor Attack for Federated Learning."* The 26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing, ACM MobiHoc (2025).

[9] **Zikai Zhang**, Ping Liu, Jiahao Xu, and Rui Hu. *"Fed-HeLLo: Efficient Federated Foundation Model Fine-Tuning with Heterogeneous LoRA Allocation."* IEEE Transactions on Neural Networks and Learning Systems, TNNLS (2025).

[10] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Detecting Backdoor Attacks in Federated Learning via Direction Alignment Inspection."* Proceedings of the Computer Vision and Pattern Recognition Conference, CVPR (2025).

[11] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Achieving Byzantine-Resilient Federated Learning via Layer-Adaptive Sparsified Model Aggregation."*, IEEE/CVF Winter Conference on Applications of Computer Vision ,WACV (2025).

[12] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Backdoored Model in Federated Learning via Individual Unlearning"*, IEEE/CVF Winter Conference on Applications of Computer Vision ,WACV (2025).

[13] **Zikai Zhang**, and Rui Hu. *"Byzantine-robust Federated Learning with Variance Reduction and Differential Privacy."* 2023 IEEE Conference on Communications and Network Security, IEEE CNS (2023).

[14] Zichen Liang, Hu Cao, Chu Yang, **Zikai Zhang**, and Guang Chen. *"Global-local Feature Aggregation for Event-based Object Detection on Eventkitti."* 2022 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, IEEE MFI (2022).

[15] Siyuan Cheng, Binfei Chu, Bineng Zhong, **Zikai Zhang**, Xin Liu, Zhenjun Tang, and Xianxian Li. *"DRNet: Towards Fast, Accurate and Practical Dish Recognition."* Science China-Technological Sciences (2021).

[16] **Zikai Zhang**, Bineng Zhong, Shengping Zhang, Zhenjun Tang, Xin Liu, and Zhaoxiang Zhang. *"Distractor-Aware Fast Tracking via Dynamic Convolutions and MOT Philosophy."* Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR (2021).

Patents:

[1] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Qihua Liang, and Xianxian Li. *"Wheat Head Detection Method based on Computer Vision Semi-supervised Pseudo Label Learning."* CN PATENT (CN113554627A, 2021).

[2] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Zhenjun Tang, Xianxian Li, Xin Liu. *"Low-power Consumption Real-time Helmet Detection Method based on Computer Vision Target Detection."* CN PATENT (CN113128476A, 2021).

PROFESSIONAL SERVICE

TPC/PC Member:

The Annual AAAI Conference on Artificial Intelligence (AAAI) 2025, 2026

International Conference on Smart Mechatronics (ICSMech) 2024

International Conference on Intelligent Knowledge Systems and Engineering Applications (IKSEA) 2024

Journal Reviewer:

IEEE Transactions on Neural Networks and Learning Systems

IEEE Transactions on Dependable and Secure Computing

IEEE/ACM Transactions on Networking

Information Processing and Management

IEEE Open Journal of the Communications Society

China Communication

Journal of Information Security and Applications

IEEE Latin America Transactions

Cluster Computing

Engineering, Technology & Applied Science Research

Conference Reviewer:

The International Conference on Learning Representations (ICLR) 2026

International Conference on Machine Learning (ICML) 2025

Conference on Neural Information Processing Systems (NeurIPS) 2022, 2024, 2025

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025, 2026

International Conference on Computer Vision (ICCV) 2025

The European Conference on Computer Vision (ECCV) 2026

Conference on Language Modeling (COLM) 2026

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025

International Conference on Pattern Recognition (ICPR) 2025

IEEE Consumer Communications & Networking Conference (CCNC) 2024, 2025

IEEE International Conference on Computer Communications and Networks (ICCCN) 2025